

High Level Programming Extensions – Procedures

Problem 1:

Create a simple procedure to get all the records from the table 'student_info'.

```
mysql> create database collegedb;
Query OK, 1 row affected (0.08 sec)

mysql> use collegedb;
Database changed
mysql> create table student_info( id int, name varchar(30), course varchar(20), marks int );
Query OK, 0 rows affected (0.14 sec)

mysql> insert into student_info values (1,'sonu','CSE',85), (2,'pranathi','ECE',90), (3,'roops','EEE',78), (4,'abhighna','CSE',88);
Query OK, 4 rows affected (0.14 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> select * from student_info;
+----+-----+-----+-----+
| id | name   | course | marks |
+----+-----+-----+-----+
| 1  | sonu   | CSE    | 85    |
| 2  | pranathi | ECE    | 90    |
| 3  | roops  | EEE    | 78    |
| 4  | abhighna | CSE    | 88    |
+----+-----+-----+-----+
4 rows in set (0.09 sec)

mysql> delimiter //
mysql> create procedure get_students() begin select * from student_info; end //
Query OK, 0 rows affected (0.09 sec)

mysql> delimiter ;
mysql> call get_students();
+----+-----+-----+-----+
| id | name   | course | marks |
+----+-----+-----+-----+
| 1  | sonu   | CSE    | 85    |
| 2  | pranathi | ECE    | 90    |
| 3  | roops  | EEE    | 78    |
| 4  | abhighna | CSE    | 88    |
+----+-----+-----+-----+
4 rows in set (0.06 sec)

Query OK, 0 rows affected (0.06 sec)
```

Problem 2:

Create a stored procedure GetCustomerLevel() that accepts two parameters customer number and customer level.

- First, it gets the credit limit from the customers table.
- Then, based on the credit limit, it determines the customer level: PLATINUM, GOLD, and SILVER.
- The parameter p_customerlevel stores the level of the customer and is used by the calling program.

```
mysql> create database collegedb;
Query OK, 1 row affected (0.02 sec)

mysql> use collegedb;
Database changed
mysql> create table customers( customerNumber int primary key, customerName varchar(30), creditLimit decimal(10,2) );
Query OK, 0 rows affected (0.14 sec)

mysql> desc customers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| customerNumber | int           | NO   | PRI | NULL    |       |
| customerName   | varchar(30)   | YES  |     | NULL    |       |
| creditLimit    | decimal(10,2) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.07 sec)

mysql> insert into customers values (101,'sonu',75000), (102,'pranathi',45000), (103,'roops',8000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select*from customers;
+-----+-----+-----+
| customerNumber | customerName | creditLimit |
+-----+-----+-----+
| 101 | sonu | 75000.00 |
| 102 | pranathi | 45000.00 |
| 103 | roops | 8000.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> delimiter //
mysql> create procedure GetCustomerLevel( in p_customernumber int, out p_customerlevel varchar(20) ) begin declare v_creditlimit decimal(10,2); select creditLimit into v_creditlimit from customers where customerNumber = p_customernumber; if v_creditlimit > 60000 then set p_customerlevel = 'PLATINUM'; elseif v_creditlimit >= 10000 then set p_customerlevel = 'GOLD'; else set p_customerlevel = 'SILVER'; end if; end //
Query OK, 0 rows affected (0.05 sec)

mysql> delimiter ;
mysql> set@level='';
Query OK, 0 rows affected (0.04 sec)

mysql> call getcustomerlevel(101,@level);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> call getcustomerlevel(101,@level);
Query OK, 1 row affected (0.00 sec)

mysql> select@level;
+-----+
| @level |
+-----+
| PLATINUM |
+-----+
1 row in set (0.00 sec)

mysql> set@level='';
Query OK, 0 rows affected (0.00 sec)

mysql> call getcustomerlevel(102,@level);
Query OK, 1 row affected (0.05 sec)

mysql> select@level;
+-----+
| @level |
+-----+
| GOLD |
+-----+
1 row in set (0.00 sec)

mysql> set@level='';
Query OK, 0 rows affected (0.00 sec)

mysql> call getcustomerlevel(103,@level);
Query OK, 1 row affected (0.00 sec)

mysql> select@level;
+-----+
| @level |
+-----+
| SILVER |
+-----+
1 row in set (0.00 sec)
```